



XMBA 200S: Data Analysis for Management

Chapter 16: Statistical Tests

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Course overview:

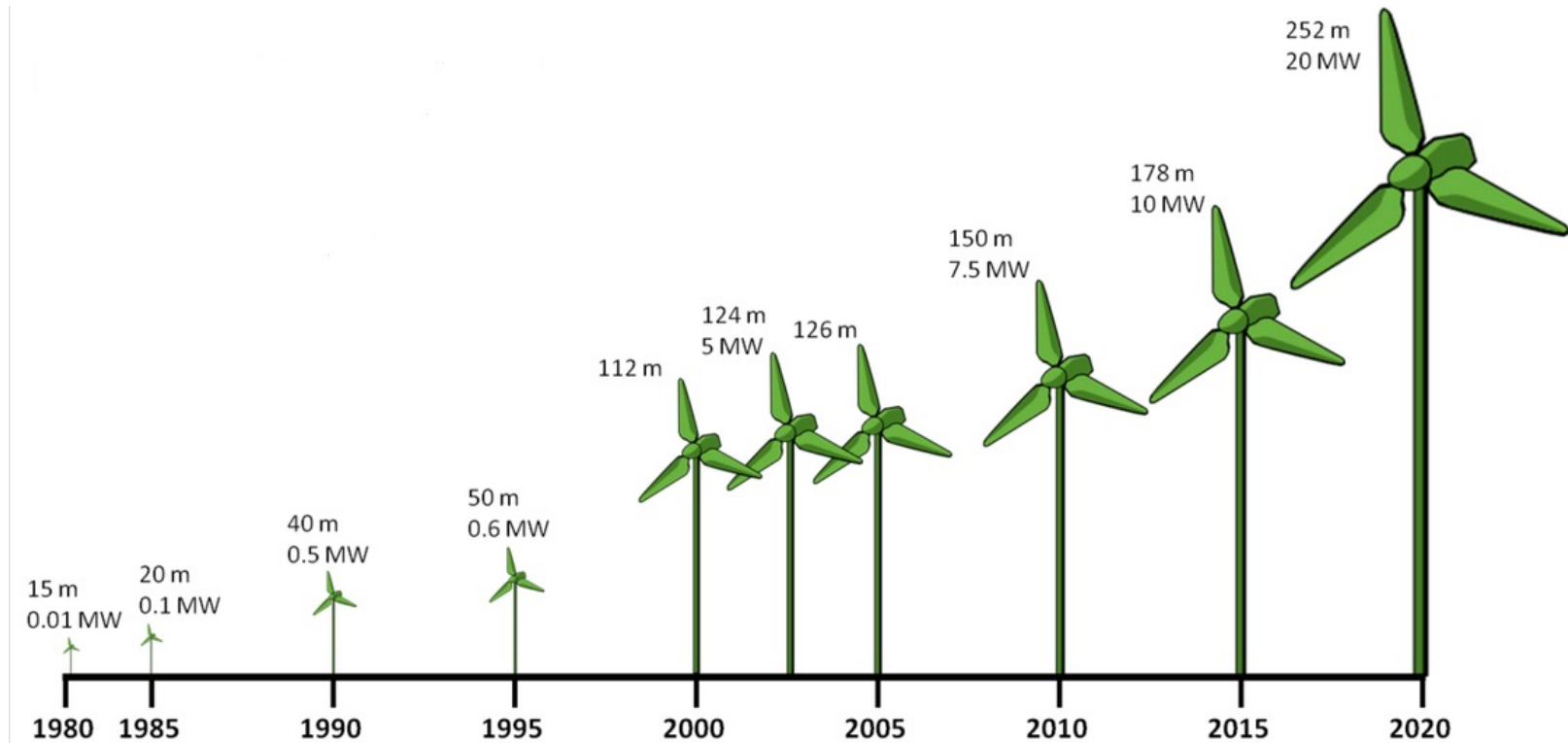
Part 2: Making decisions based on inference

Given **limited data**, managers need to choose how to act

- Typically, you'll have some benchmark in mind, or two (or more) groups for comparison
- But how do you know any difference isn't due to "noise"?
- What is the likelihood of making a mistake?

In the next two classes, we build on what we now know about distributions of sample statistics to conduct **statistical tests** that provide us with some clarity

Wind Turbines Profitability



- U.S. electricity generation from wind has increased 10-fold since 2005.
- Part of the explanation for the increase is turbine size.
- Today many turbines have a rotor diameter that exceeds 120 meters.
- Turbine blades must be light weight and extremely strong.

Wind Turbines

Nordex is experimenting with a new production process based on carbon fiber

- Production is more costly, but blades are lighter and more efficient
- The finance team told you that the switch would be profitable for the company if the mean weight of new tech is less than **29.7 tons**.
- You are interested in seeing whether you can manufacture new blades that are light enough to justify the switch
- **Run a Pilot:** Under carefully controlled conditions, you manufacture $n = 40$ carbon fiber blades as a sample.
- Based on the results of this pilot, should you switch to the new material? Data and Decisions...

Null and Alternative Hypotheses

Your decision depends on what the new tech parameter μ is.

The **null hypothesis (H_0)** specifies the default course of action, preserving the status quo (i.e. no change).

The **alternative hypothesis (H_a)** contradicts the assertion of the null hypothesis, posits an alternative to the status quo (i.e. change to new tech)

The null & alternative hypotheses are claims about a population parameter (in this case, weight of turbine blades).

Wind Turbines

Let μ denote the weight of turbine blades.

$H_0:$

$H_a:$

These hypotheses lead to a _____ test. Why?



One and Two-Sided Tests

With a **one-sided test** the null hypothesis allows any value of a parameter larger [$>$] (or smaller [$<$]) than a specific value

With a **two-sided test** the null hypothesis asserts a specific value ($=$) for the population parameter.

Which you use depends on the business problem.

- When a problem calls for a two-sided test, it will often make sense to report the confidence interval instead.

Type I and Type II Errors

		Company's Decision	
		Don't Take Action Retain H_0	Take Action Reject H_0
State of the World	$H_0: \mu \geq 29.7$	Correct Decision	Type I Error
	$H_a: \mu < 29.7$	Type II Error	Correct Decision

Statistical Tests

Test to quantify how unusual it is to observe a particular value of a statistic under the null hypothesis.

- If it is unusual, then we “reject the null hypothesis”.
- If it is not unusual, then we “fail to reject the null hypothesis”.

Reminder:

- Our statistics are random variables (“noisy”)
- They come from a distribution that is approximately normal with a standard error that we can calculate.

If I give you a number, you have the tools to quantify how unlikely that number would be under the null

Some Terminology: Statistical Significance

Significance level (α): the maximum tolerance for a Type I error (typically $\alpha = 0.05$ or $\alpha = 0.01$)

Statistically significant: the data contradict the null hypothesis at our chosen significance level. We “reject the null hypothesis”

Suppose in the Nordex example, you decide that $\alpha = 0.05$ is our maximum tolerance for a Type I error.

- For what values of $\bar{X}$ would you reject the null hypothesis?
- To answer this question, it's helpful to calculate a **p-value**

Some Terminology: The p-value

The **p-value** is the smallest significance level at which we would reject the null hypothesis.

A low p-value is evidence against the null hypothesis.

We say a statistic is **statistically significant** if the p-value is less than our adopted significance level (α).

- If a statistic is statistically significant: “reject the null hypothesis”

Using Data to Make a Decision

A statistical test is a decision rule to separate H_0 from H_a

1. Define your Type I error α (typically 5% or 1%). α is called the **significance level** of the test here
2. Conduct Pilot, Collect Sample Data
3. Construct a **test statistic** $t = \frac{\bar{X} - \mu_0}{SE}$
4. Find the p-value associated with that t-statistic
 - Hint: use the t-table to find the corresponding probability
5. Reject the null hypothesis H_0 if p-value $< \alpha$

Wind Turbines: Pilot Results

Pilot: you build 40 carbon turbine blades with a **mean ($\bar{x}$) of 29.5** and a **standard deviation (s) of 0.7**

Do we reject null based on a significance level $\alpha = .05$?

What is the *smallest* α for which we would have rejected?
(i.e. what is the **p-value**?)

In a world where the null is true, how likely is it to see a statistic as small or smaller than 29.5?

The p-value

Calculating the p-value of our statistic:

$$\text{Probability}(\bar{x} \leq 29.5)$$

Very similar to before, but we need to know how to “standardize” our statistic (i.e. subtract the mean and divide by the standard error)

- Which parameter/mean to subtract?
- How do we calculate the standard error?
- Hint: “In a world where the null is true”

With a one-sided test, the null has an infinite number of values...

- We use the parameter value under the null that gives the largest probability of a Type I error ($\mu = 29.7$)

The p-value

Calculating the p-value of our statistic:

$$\text{Probability}(\bar{x} \leq 29.5)$$

Under the null hypothesis $\bar{x}$ has a **standard error** _____. Why?

Subtracting off the mean (under the null) and dividing by the standard error gives us a Student **t-distribution** (df=____)

On the t-table, this corresponds to _____

Thus, our p-value is _____

This is the smallest α at which we would reject the null hypothesis

Wind Turbines Test

1. We decided to adopt a significance level of $\alpha = 0.05$
2. We constructed the t statistic for our problem:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{29.5 - 29.7}{0.1107} = -1.807$$

3. You could look this up in Excel (or t-table) with $df = 40 - 1 = 39$ & find a p-value = $T.DIST(-1.807, 39, TRUE) = 0.039$
4. Do you adopt the new process? Yes, because $0.039 < 0.05$
(Reject the Null)

What About Claude?

you build 40 carbon turbine blades with a mean of 29.5 and a standard deviation (s) of 0.7. Do we reject null (29.7) based on a significance level $\alpha = .05$ with a one sided test?

Setting up a one-sided hypothesis test for a sample mean.

Yes, reject the null, assuming the alternative is the lower-tail one ($H_a: \mu < 29.7$), which is the direction the sample mean points.

- Standard error: $0.7 / \sqrt{40} \approx 0.1107$
- Test statistic: $t = (29.5 - 29.7) / 0.1107 \approx -1.81$
- Critical value ($df = 39, \alpha = .05$, one-sided): -1.685 (or -1.645 using z)
- p-value: ≈ 0.039 (≈ 0.035 using z)

Since $-1.81 < -1.685$, you reject H_0 at the 5% level.

The result depends on the one-sided setup. A two-sided test would have $p \approx 0.078$ and would fail to reject. If the alternative were instead $\mu > 29.7$, you would also fail to reject.

What Does the P-Value Tell You?

The decision rule using your **p-value** is simple:

If $\text{p-value} < \alpha$, you reject the Null

If $\text{p-value} \geq \alpha$, you fail to reject the Null

- Not enough evidence to justify change, so keep status quo
- The sample you saw could happen frequently enough under the Null
- More technically: given what you see in your data, the probability of a Type I error is too high

Accommodating Two-Sided Hypotheses

Sometimes you may want to test a “two-sided” hypothesis.

- Is the new process different from the old one?

Two-sided hypothesis tests are set up as follows:

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

The math is the same, except for what rule you use:

- In two-sided tests, you need:
p-value_twosided = 2 * p-value_onesided < α to reject the null
(i.e. significance level is split across the two tails of the distribution).
- When $\alpha = 0.05$, you allocate 0.025 to each tail... but still compare the test statistic p-value directly to 0.05 for rejecting null.

Note: What Standard Error to Use for a Statistical Test?

If your variable can take on any value, you should use the sample standard deviation in the numerator

$$SE = \frac{s}{\sqrt{n}}$$

If your variable only takes on the values of 0 or 1, use the following formula where p_0 is the proportion specified by the null hypothesis.

$$SE = \frac{\sqrt{p_0(1 - p_0)}}{\sqrt{n}}$$

Note 1: here p_0 is **not** the “p-value”!

Note 2: Statistical software will choose the right SE for you!

Decision Rules and Probability of a Type II Error

When we decide upon a significance level (α), we can solve for the corresponding value of $\bar{x}$ that is consistent with that significance level. We call this an action threshold or decision rule...

Find a **decision rule that satisfies:**

$$Probability(\bar{x} \leq \text{decision rule}) = 0.05$$

Under the null $\bar{x}$ **has a t-distribution** (df=39) with mean 29.7 and **standard error** $\frac{s}{\sqrt{n}} = 0.1107$

Standardizing our statistic gives us a Student t-distribution.

$$Probability\left(t \leq \frac{\text{decision rule} - 29.7}{0.1107}\right) = 0.05$$

The **t-table** allows us to move between probabilities and t-statistics... for example, the table tells us that **5% of observations** lie to the **left of -1.691**. Thus, we can solve for a decision rule that yields a t-statistic of -1.691

$$\frac{\text{decision rule} - 29.7}{0.1107} = -1.691$$

We conclude that with $\alpha = .05$ our **decision rule is to reject** if $\bar{x} \leq 29.51$

Wind Turbines – Type II Error

- With $\alpha = 0.05$, decision rule to **reject null if** $\bar{x} \leq 29.51$
- **Scenario**: Suppose that the true weight (μ) is 29.3

Given our decision rule, what is the probability of a Type II error?



Wind Turbines - Type II Error

Given $\mu = 29.3$ and the company's decision rule, what is the probability of a Type II error?

That is, under this alternative hypothesis **there is a _____ chance that the company will commit a Type II error**

Type II Error and Power

The **power** of a test is 1 minus the probability of a Type II error.

For example, in the Nordex example we can say that the power of the test when $\mu = 29.3$ is 95% Why? $1 - 0.05 = 0.95$.



Type II Errors and Power

To calculate the probability of a Type II error, you need a hypothesized scenario under the alternative hypothesis

Type II Error probability depends upon a chosen scenario for what the alternative hypothesis is!

Designing statistical tests:

- Often, we choose a sample size for a study such that, under alternatives you think are reasonable, you have a specific chance of being able to reject the null.
- This is called doing “**power calculations**” and is an essential input into the design of pilots or experiments.
- Conventionally, we set $\alpha = 0.05$ and power at 80% and solve for n

See more on bCourses!

Other Properties of Tests

Significance versus Importance

- Statistical significance does not mean that you have made an important or meaningful discovery.
- The size of the sample affects the p-value of a test.
 - With enough data, a trivial difference from H_0 leads to a statistically significant outcome.

Other Properties of Tests

Confidence Interval or Test?

- A confidence interval provides a range of parameter values that are compatible with the observed data.
- A test provides a precise analysis of a specific hypothesized value for a parameter.

Pitfalls

- Do not confuse statistical significance with substantive importance.
- Do not think that the p -value is the probability that the null hypothesis is true.
- Avoid cluttering a test summary with jargon.

Takeaways

- Pick the hypotheses before looking at the data.
- Choose the null hypothesis on the basis of profitability.
- Pick the α level first, taking into account both types of error.
- Think about whether $\alpha = 0.05$ is appropriate for each test.
- Report a p -value to summarize the outcome of a test.